AWS SAA-C03 Study Guide: Disaster Recovery & Migrations

1. Disaster Recovery (DR) in AWS

The SAA-C03 exam focuses heavily on the trade-off between **RTO/RPO** and **Cost**.

- **RPO (Recovery Point Objective):** Maximum acceptable data loss (measured in time, e.g., "1 hour of data").
- **RTO (Recovery Time Objective):** Maximum acceptable downtime (how long until services are back up).

The 4 DR Strategies (From Cheapest/Slowest to Expensive/Fastest)

Strategy	Description	RTO/RPO	Cost
Backup & Restore	Back up data to S3 or Snapshots. Restore only when disaster strikes. Infrastructure is provisioned *after* the disaster.	High (Hours/Days)	$ (Lowest)
Pilot Light	Core data is replicated (e.g., RDS RR), but app servers are off or very small. "Turn on" and scale up servers during disaster.	Medium (Tens of minutes)	$$
Warm Standby	A scaled-down version of the full environment is always running (e.g., min capacity Autoscaling Group). Scale up on failover.	Low (Minutes)	$$$
Multi-Site (Active/Active)	Full capacity running in multiple regions/AZs simultaneously. Traffic is load balanced (Route53).	Near Zero	(Highest)

2. Elastic Disaster Recovery (AWS DRS)

- **Purpose:** The standard AWS service for Disaster Recovery (replaces CloudEndure DR).
- **How it works:**
  1. Install the DRS Agent on source servers (On-prem or other clouds).
  2. Continuous block-level replication to a **Staging Area** in AWS (low-cost, lightweight EC2 instances).
  3. **Failover:** When triggered, DRS launches fully provisioned EC2 instances based on the replicated data.

- **Exam Keyword:** "Continuous replication," "Block-level," "Physical-to-Cloud DR," "Sub-second RPO."

3. Database Migration Service (DMS)

- **Purpose:** Migrate databases to AWS quickly and securely.
- **Key Feature:** The source database remains **fully operational** during the migration.
- **Homogeneous Migrations:** Same engine (e.g., Oracle to Oracle on EC2/RDS). No schema conversion needed.
- **Heterogeneous Migrations:** Different engine (e.g., Oracle to Aurora PostgreSQL). Requires **SCT**.
- **AWS Schema Conversion Tool (SCT):** Must run *before* DMS to convert the schema (tables, indexes, views) of the source engine to the target engine.
- **CDC (Change Data Capture):** Enable this in DMS for continuous replication after the initial load.

4. RDS & Aurora Migrations

RDS

- **Snapshot & Restore:** Standard way to move data or create environments.
- **Read Replicas:**
  - Can be within the same region or **Cross-Region**.
  - **DR Strategy:** Create a Cross-Region Read Replica. If the primary region fails, **Promote** the replica to be the standalone primary (RTO is minutes).

Aurora

- **Aurora Global Database:**
  - Spans multiple AWS regions.
  - Physical replication (very fast, < 1 second latency).
  - **Fast Failover:** If primary region goes down, promote secondary region (RTO < 1 minute).
- **Aurora Cloning:**
  - Create a new cluster from an existing one almost instantly.
  - **Use Case:** Testing/Staging. *Not* typically used for DR because it resides in the same region.

⚡ Specific Migration Scenarios (Exam Favorites)

1. RDS MySQL -> Aurora MySQL (or RDS Postgres -> Aurora Postgres)

- **Option A: Aurora Read Replica (Recommended for Min Downtime)**
  1. Create an **Aurora Read Replica** from the source RDS instance.

2. Wait for replication lag to drop to 0.
3. Stop writes to the RDS instance.
4. **Promote** the Aurora Read Replica to a standalone Aurora Cluster.

- **Option B: Migrate from Snapshot**
  - Create a snapshot of RDS -> "Migrate Snapshot" to Aurora.
  - **Cons:** Slower, requires downtime to restore data to a new cluster.

2. External (On-Prem/EC2) -> Aurora

- **External MySQL -> Aurora MySQL**
  - **Percona XtraBackup (Physical):** Create a physical backup using Percona XtraBackup -> Upload to **S3** -> Restore directly to Aurora. (Much faster than logical dumps).
  - **mysqldump (Logical):** Standard SQL dump (Slower for large datasets).
- **External PostgreSQL -> Aurora PostgreSQL**
  - **Import from S3:** Use `pg_dump` to export data -> Upload to **S3** -> Use the `aws_s3` Aurora extension to import data.
  - **DMS:** Use for continuous replication and minimal downtime cutover.

5. On-Premises Strategies with AWS (Hybrid)

VM Management & Migration Planning

- **Amazon Linux 2 AMI:**
  - You can download the Amazon Linux 2 AMI as a VM image (`.iso` format).
  - Runs on: VMWare, KVM, VirtualBox (Oracle VM), and Microsoft Hyper-V.
  - **Use Case:** Develop and test applications locally on the exact same OS used in AWS.
- **VM Import/Export:**
  - **Migrate:** Import existing VM images into EC2 instances.
  - **DR Strategy:** Create a DR repository for your on-premise VMs in AWS.
  - **Export:** You can export instances back from EC2 to your on-premises virtualization environment.
- **AWS Application Discovery Service:**
  - **Purpose:** Gather information about on-premise servers to plan a migration.
  - **Data Collected:** Server utilization (CPU/RAM) and **dependency mappings** (which server talks to which).
  - **Tracking:** Integrates with **AWS Migration Hub** to track the status of migrations.
- **Strategic Migration Services (On-Prem Context):**
  - **DMS:** Replicate On-prem => AWS, AWS => AWS, or AWS => On-prem. Works with various DBs (Oracle, MySQL, DynamoDB, etc.).

- **MGN (Application Migration Service):** Incremental replication of on-premises live servers to AWS.

Storage Strategies

1. **AWS Storage Gateway** Connects on-prem applications to cloud storage.
   - **S3 File Gateway:**
     - Interface: NFS or SMB.
     - Backend: S3.
     - Use Case: Moving files to S3, content repositories. Local cache keeps frequently accessed files fast.
   - **Volume Gateway:**
     - Interface: iSCSI (Block storage).
     - **Stored Mode:** All data stays on-prem; async backup to S3 (Good for low latency access to *all* data).
     - **Cached Mode:** Primary data is in S3; frequently accessed data is cached on-prem (Good for storage extension).
   - **Tape Gateway:**
     - Interface: Virtual Tape Library (VTL).
     - Backend: S3 Glacier / Deep Archive.
     - Use Case: Replacing physical tape backups (i.e., BackupExec, Veeam).
2. **AWS DataSync**
   - **Purpose:** Automate moving large amounts of data between on-prem and AWS storage (S3, EFS, FSx).
   - **Tech:** Uses a proprietary protocol (not standard NFS/SMB over WAN) which is up to **10x faster** than open-source tools.
   - **Agent:** Requires a DataSync agent on-prem.

6. AWS Backup

- **Purpose:** Centralized, managed service to automate backups across AWS services.
- **Supported Services:** EC2, EBS, RDS, DynamoDB, Aurora, EFS, FSx, Storage Gateway, S3.
- **Key Features:**
  - **Backup Plans:** Define frequency (cron) and retention.
  - **Cross-Region Backup:** automatically copy backups to another region for DR.
  - **Cross-Account Backup:** Managed via AWS Organizations.
  - **Vault Lock:** Enforces **WORM** (Write Once Read Many) compliance. Prevents deletion of backups (protection against Ransomware).

7. Application Migration Service (MGN)

- **Purpose:** The primary "Lift and Shift" (Rehost) tool for migrating servers to AWS.
- **Replaces:** Server Migration Service (SMS) and CloudEndure Migration.
- **How it works:**
  - Agent-based block-level replication.
  - Converts source servers (Physical, VMware, Hyper-V, Azure VMs) to native AWS EC2 instances.
- **Cutover:** Minimal downtime. You define a "Cutover window."

8. Transferring Large Datasets (The "Snow" Family)

Rule of Thumb: If data transfer over the internet/Direct Connect takes > 1 week, use Snowball.

Device	Capacity	Use Case	Compute Capability?
Snowcone	8 TB	Small, portable, harsh environments. Can be mailed.	Basic (IoT/Validation)
Snowball Edge Storage Optimized	80 TB	Massive data transfer (Petabytes scale).	Standard EC2/Lambda
Snowball Edge Compute Optimized	42 TB	Edge computing (AI/ML inference at the edge) + Transfer.	Powerful EC2/GPU
Snowmobile	100 PB	Exabyte scale data transfer (Literal truck).	No

- **OpsHub:** The GUI software used to manage Snow devices locally.
- **Direct Connect (DX):** Dedicated private fiber connection. Good for consistent high throughput, but takes time to set up (months). Use DX for *ongoing* hybrid connectivity, use Snowball for *one-time* massive migration.

9. VMware Cloud on AWS

- **Concept:** Run the full VMware software stack (vSphere, vSAN, NSX) on AWS **Bare Metal** infrastructure.
- **Key Use Case:**
  - Rapid migration without re-architecting applications.
  - Keep using existing VMware tools (vCenter) to manage AWS resources.
- **vMotion:** You can live migrate (vMotion) VMs from on-prem data centers to VMware Cloud on AWS with **zero downtime**.
- **Integration:** Can directly access native AWS services (S3, RDS, etc.) via the Elastic Network Interface (ENI).

Quick Exam Cheat Codes

1. **Heterogeneous Database Migration? -> DMS + SCT**.
2. **Lift and Shift Servers? -> Application Migration Service (MGN)**.
3. **Physical Tapes? -> Tape Gateway**.
4. **NFS/SMB to S3? -> File Gateway**.
5. **iSCSI Block Storage? -> Volume Gateway**.
6. **Need < 15 min RPO/RTO but save money? -> Pilot Light** (or Warm Standby if RTO needs to be minutes).
7. **Transfer 500 TB where internet is slow? -> Snowball Edge**.
8. **Prevention against accidental backup deletion? -> AWS Backup Vault Lock**.
9. **External MySQL to Aurora Migration? -> Percona XtraBackup to S3**.